SRIKAR YALAMARTHI

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Education

Vellore Institute of Technology, Vellore — Sep 2022 – Jul 2027
Integrated M.Tech in Computer Science and Engineering — CGPA: 8.49

Dr. K.K.R's Happy Valley School — 2021 – 2022
Class XII — Aggregate: 90.4%

Akshara School — 2019 – 2020
Class X — Aggregate: 80%

Projects

Crop Connect – AI-Integrated Smart Agriculture System

- Developed a full-stack smart agriculture platform integrating AI-based crop disease detection and weather forecasting.
- Built secure REST APIs with optimized backend processing and efficient database management.
- Designed a scalable modular architecture supporting e-commerce, crop analytics, and AI service integration.

Multi-Hazard Prediction System Using Adaptive Terrain Graph

- Developed a Digital Twin-based system for real-time flood and landslide risk prediction using terrain and environmental data.
- Designed a graph-based terrain model to simulate hydrological flow and hazard propagation.
- Implemented adaptive spatial refinement to improve prediction while optimizing computational efficiency.
- Built a hybrid Physics + XGBoost prediction engine for real-time hazard probability mapping.

Caption Craft – Hybrid Speech Recognition & RAG-Based Transcript Intelligence

- Developed an Indian-accented English speech recognition system using Bi-LSTM + CTC with Mel-spectrogram features and Whisper-based hybrid transcription.
- Built a Groq LLM pipeline for transcript summarization and extraction of key points, decisions, and actions.
- Implemented RAG-based transcript Q&A using LangChain, Sentence Transformers, and ChromaDB for context-grounded responses.

Patents

Multi-Hazard Prediction System Using Adaptive Terrain Graph

- Published Indian Patent Application, June 2026 (Patent Application No. 202641065384).
- Proposed an adaptive terrain graph-based framework for multi-hazard and environmental risk prediction.

Technical Skills

Languages & Development: Python, Java(OOPS), SQL

AI, ML & GenAI: Machine Learning, NLP, LLMs, RAG, LangChain, MCP, ChromaDB, Data Cleaning & Preprocessing

Databases & Tools: MySQL, MongoDB, Git/GitHub, Docker, FastAPI, REST APIs, Postman

Core Concepts: DSA, Operating Systems, DBMS, Computer Networks, Cyber Security

Certifications

- Microsoft Azure AI Fundamentals – Microsoft — Credential
- AI Upskilling Certificate: Technical Foundations – Qualcomm Technologies, Inc. — Credential
- Networking Basics – Cisco Networking Academy — Credential
- Introduction to Cybersecurity – Cisco Networking Academy — Credential